

Question	Marking guidance	Mark	Comments
08.1	Three of chromosome 13 / an extra chromosome 13;	1	Accept trisomy 13 Accept circle around three chromosomes or any other correct indication on Figure 8 Do not allow references to any other chromosomes. Do not accept chromatids for chromosomes.
08.2	1. In meiosis; 2. Homologous chromosomes / sister chromatids do not separate;	2 max	2. Accept non-disjunction
08.3	1. Mutation / extra chromosome in gamete/egg/sperm (that formed zygote); 2. All cells derived (from a single cell/zygote) by mitosis; OR 3. All cells derived from a single cell/zygote by mitosis; 4. Mitosis produces genetically identical cells / a clone;	2	Mark points 1 and 2 OR 3 and 4 4. Accept: have same DNA / same alleles
08.4	1. (Some) oxygenated blood (from the aorta) flows into pulmonary artery; OR Less oxygenated blood flows out through aorta; OR Lower blood pressure in aorta; 2. Less oxygen delivered to cells / tissues / organs / named organ / via named blood vessel; 3. So less/not enough oxygen for <u>aerobic</u> respiration (in cell/ tissue/organ); 4. Tissue/organ doesn't grow/develop properly (causing death); OR Tissue dies/organ stops working (causing death);	3 max	1. Accept mixing of deoxygenated with oxygenated blood in pulmonary artery 2. Do not accept "no oxygen" 3. Do not accept "produce energy"